

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

General Instructions:

1. This paper has three sections: Section A (MCQs), Section B (Short Questions), and Section C (Long Questions).
2. Attempt Section A on the question paper by filling the bubble next to the correct option.
3. In Section B, attempt any 11 questions. For each, choose either the main question **OR** its alternative.
4. In Section C, attempt all 4 long questions. For each, choose either the main question **OR** its alternative.
5. Use black or blue ink only. Atomic masses: H=1, C=12, N=14, O=16, Na=23, S=32, Cl=35.5, K=39, Ca=40, Fe=56, Cu=63.5. $N_A = 6.022 \times 10^{23}$.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The branch of chemistry that deals with the study of compounds of carbon is:	Physical chemistry	Inorganic chemistry	Organic chemistry	Analytical chemistry	A B C D
2	The direct change of a substance from the solid state to the gaseous state is called:	Sublimation	Condensation	Fusion	Deposition	A B C D
3	Isotopes of an element have the:	Same neutrons, different protons	Same protons, different neutrons	Same electrons, different protons	Same mass number, different atomic number	A B C D
4	The maximum number of electrons that the third shell (M, n=3) can hold is:	8	32	2	18	A B C D
5	Moving from left to right across a period, the atomic radius generally:	Decreases	Increases	Stays constant	First decreases then increases	A B C D
6	A bond formed by the complete transfer of electrons from a metal to a non-metal is a:	Covalent bond	Coordinate covalent bond	Ionic bond	Metallic bond	A B C D
7	One mole of any substance contains the following number of representative particles:	6.022×10^{22}	6.022×10^{24}	6.022×10^{20}	6.022×10^{23}	A B C D
8	The empirical formula of benzene (C_6H_6) is:	CH	C_6H_6	CH_2	C_2H_2	A B C D
9	In terms of electrons, oxidation is defined as the:	Gain of electrons	Loss of electrons	Gain of protons	Loss of protons	A B C D
10	For an exothermic reaction, the enthalpy change (ΔH) is:	Negative	Positive	Zero	Always equal to one	A B C D
11	At dynamic equilibrium, the rate of the forward reaction is ____ the rate of the reverse reaction:	Greater than	Equal to	Less than	Zero compared to	A B C D
12	The rusting of iron in the presence of air and moisture is an example of:	Reduction	Neutralization	Sublimation	Corrosion	A B C D

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Define chemistry. Name THREE branches of chemistry and state what each one studies.	3	OR	Differentiate between science, technology, and engineering with one example of each.	3
2(ii)	Differentiate between an element, a compound, and a mixture, giving one example of each.	3	OR	Differentiate between a solution, a suspension, and a colloid, giving one example of each.	3
2(iii)	Define atomic number and mass number. Calculate the number of protons, neutrons, and electrons in $^{35}_{17}\text{Cl}$.	3	OR	Define isotopes. Name the three isotopes of hydrogen and state the composition of each nucleus.	3
2(iv)	Write the electronic configuration of (a) sodium (Z=11) and (b) chlorine (Z=17), and state the number of valence electrons of each.	3	OR	State the modern periodic law. How are the elements arranged into groups and periods in the modern periodic table?	3
2(v)	Define ionization energy. State how it varies (a) down a group and (b) across a period.	3	OR	State THREE differences between metals and non-metals.	3
2(vi)	Define ionic bond and covalent bond. Give one example of each.	3	OR	Using electron distribution, explain the formation of magnesium chloride (MgCl_2) from its atoms.	3
2(vii)	Calculate the molecular mass of sulphuric acid (H_2SO_4). (H=1, S=32, O=16)	3	OR	Calculate the number of moles present in 36 g of water (H_2O). (H=1, O=16)	3
2(viii)	Balance the following chemical equation: $\text{Fe}_{(\text{s})} + \text{O}_{2(\text{g})} \rightarrow \text{Fe}_2\text{O}_{3(\text{s})}$	3	OR	Define empirical formula and molecular formula. Give one example of each.	3
2(ix)	Define oxidation and reduction in terms of loss and gain of electrons. Give one example of each.	3	OR	Calculate the oxidation number of sulphur (S) in sulphuric acid (H_2SO_4). Show your working.	3
2(x)	Define exothermic and endothermic reactions, giving one example of each.	3	OR	Define activation energy. How does a catalyst affect the activation energy and the rate of a reaction?	3
2(xi)	Define a reversible reaction and dynamic equilibrium. Give one example of a reversible reaction.	3	OR	State THREE characteristics of a chemical system that has reached dynamic equilibrium.	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	(a) State the main postulates of Bohr's atomic model. [3] (b) Write the electronic configuration of potassium (Z=19) and calcium (Z=20), and identify the valence electrons of each. [2]	5	OR	(a) Compare the three subatomic particles (proton, neutron, electron) in terms of charge, relative mass, and location in the atom. [3] (b) Define atomic number, mass number, and isotope, with one example each. [2]	5
3(ii)	(a) Explain the following periodic trends across a period, with reasons: (i) atomic size (ii) ionization energy (iii) electronegativity. [3] (b) Differentiate between electropositive and electronegative elements, with one example of each. [2]	5	OR	(a) Using dot-and-cross diagrams, explain the formation of an ionic bond in NaCl and a covalent bond in Cl_2 . [3] (b) State FOUR differences between ionic and covalent compounds. [2]	5

3(iii)	(a) Define mole and Avogadro's number. [2] (b) Calculate: (i) the number of moles in 49 g of H_2SO_4 (ii) the number of molecules in 0.5 mol of CO_2 (iii) the mass of 2 moles of NaOH. (H=1, S=32, O=16, C=12, Na=23) [3]	5	OR	(a) Balance the following equations: (i) $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ (ii) $\text{Al} + \text{HCl} \rightarrow \text{AlCl}_3 + \text{H}_2$ (iii) $\text{KClO}_3 \rightarrow \text{KCl} + \text{O}_2$. [3] (b) Calculate the molecular mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and the formula mass of calcium carbonate (CaCO_3). (C=12, H=1, O=16, Ca=40) [2]	5
3(iv)	(a) Define oxidation and reduction in terms of electron transfer. In the reaction $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$, identify the oxidizing agent and the reducing agent, with reasons. [3] (b) What is corrosion? State TWO methods used to prevent the rusting of iron. [2]	5	OR	(a) Define exothermic and endothermic reactions. Draw a labelled energy diagram for each and give one example of each. [3] (b) Define dynamic equilibrium. State TWO conditions necessary for a system to attain dynamic equilibrium. [2]	5

Invigilator's Signature: _____ Candidate's Name: _____ Roll No: _____

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— End of Question Paper —